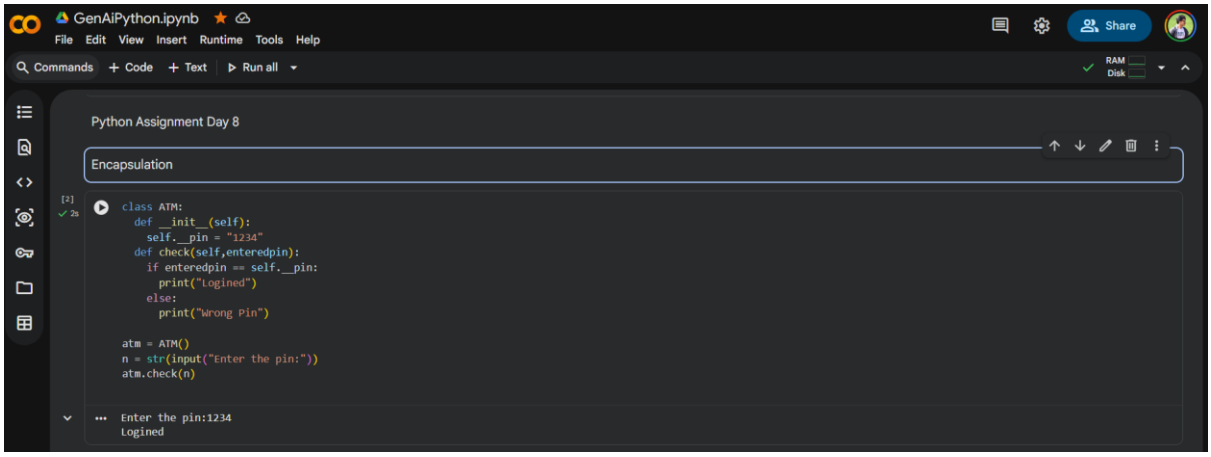
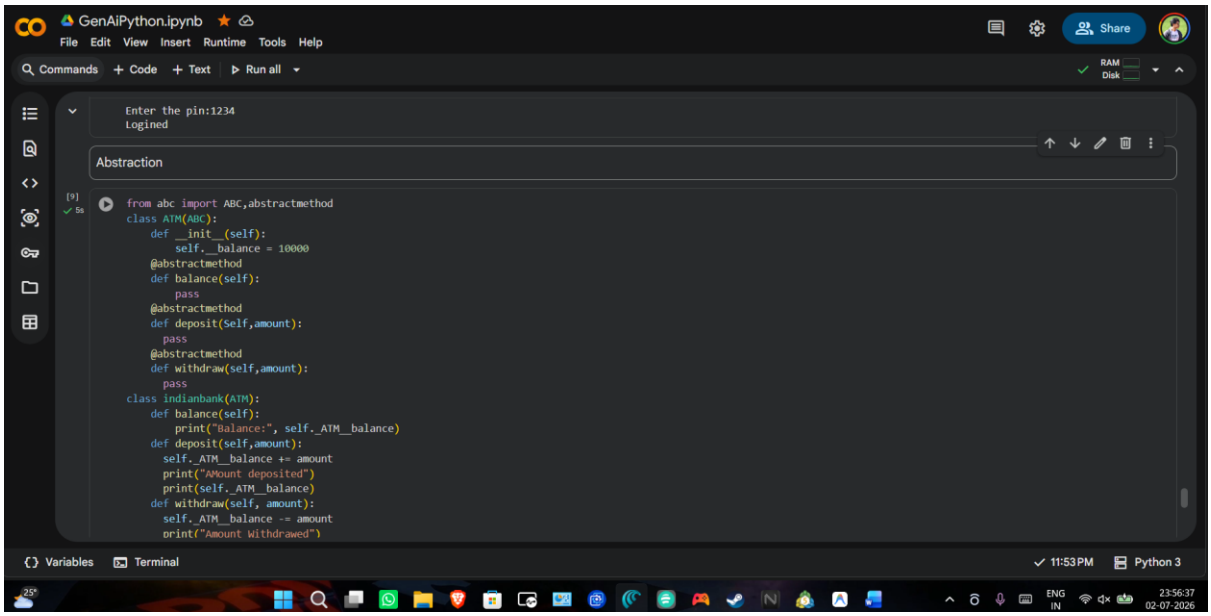
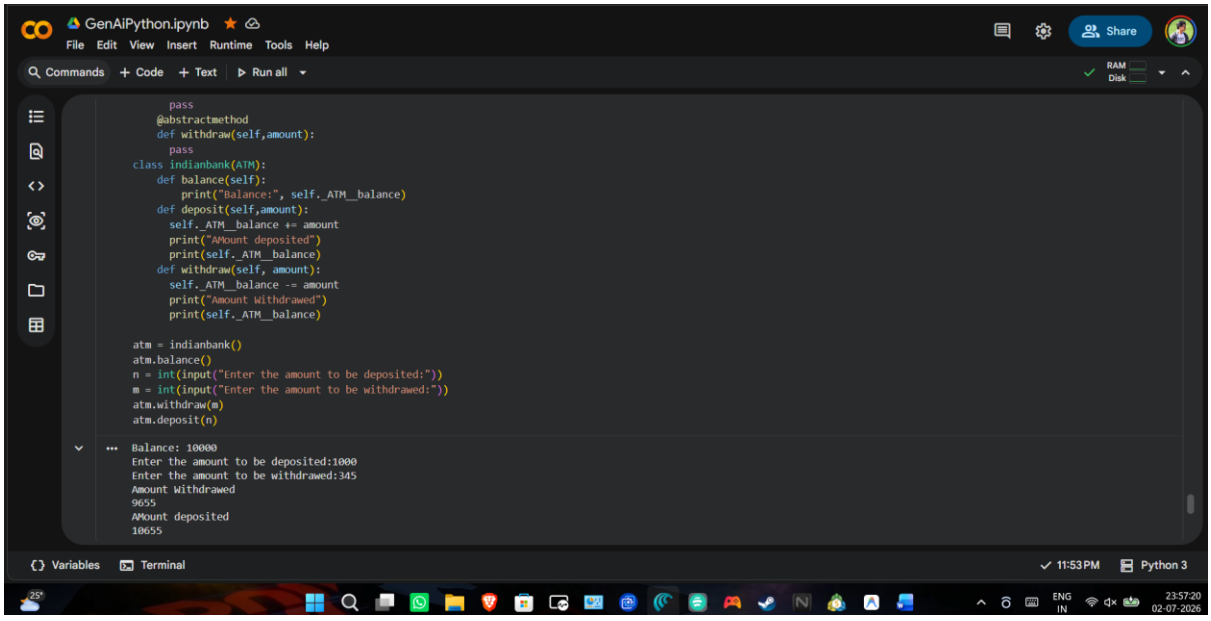
Python Assignment Day 8

1. Encapsulation

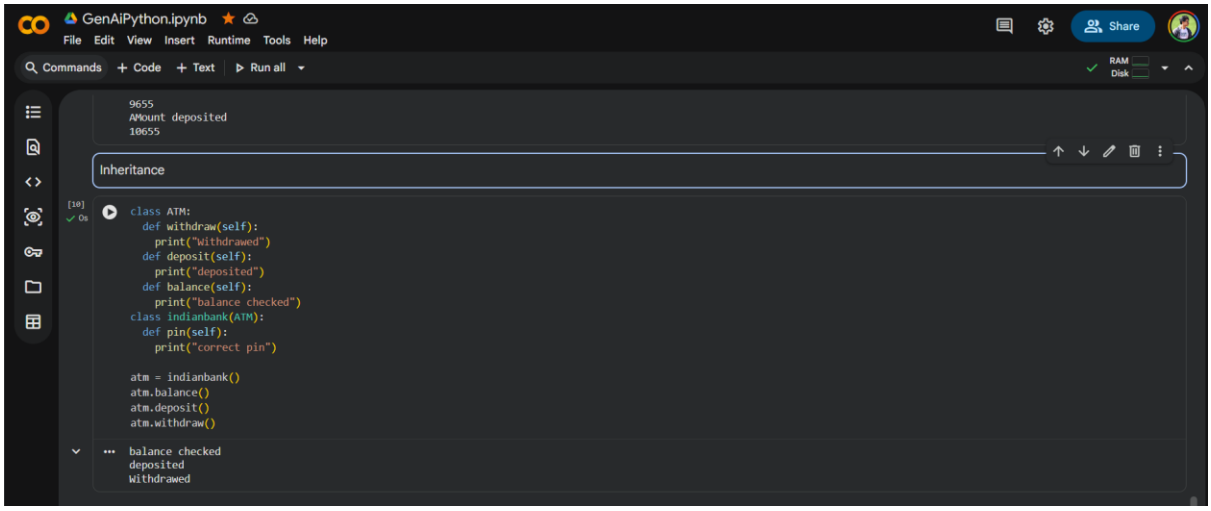


2. Abstraction

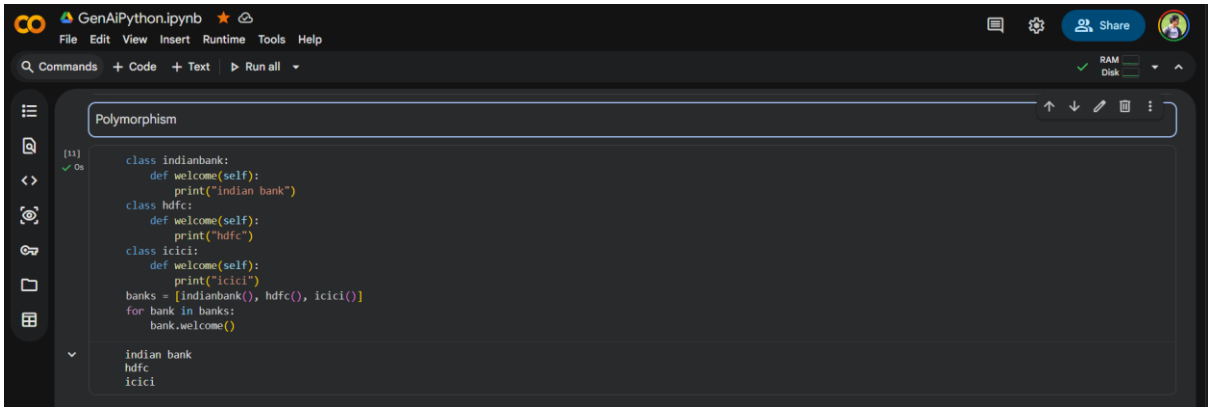




3. Inheritance

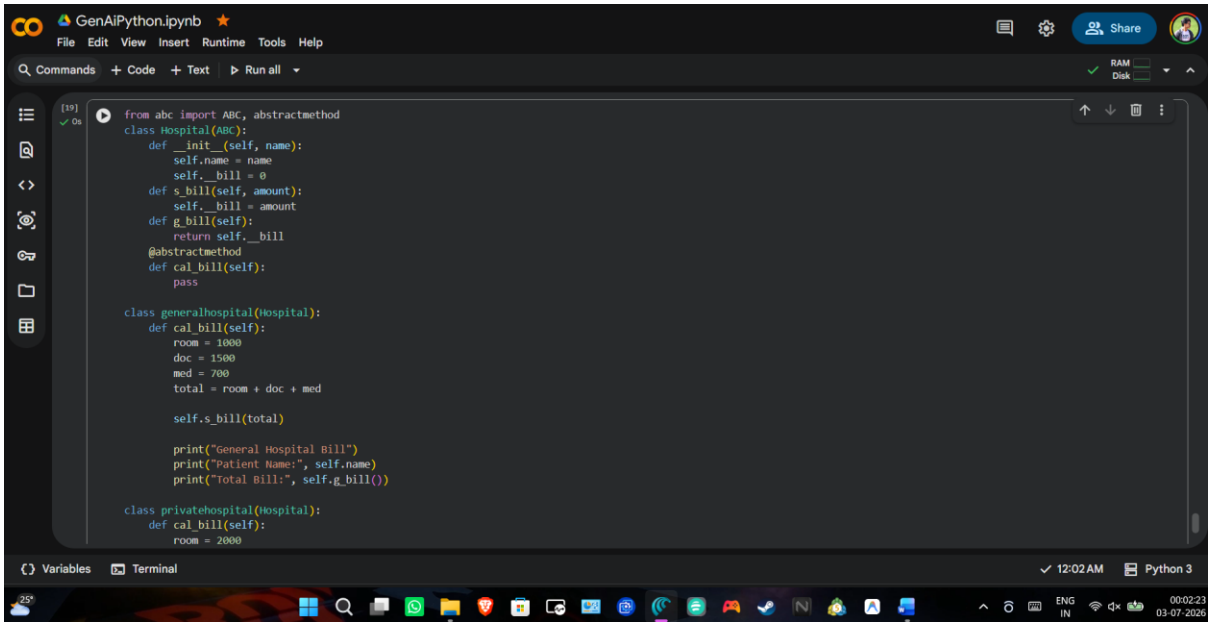


4. Polymorphism

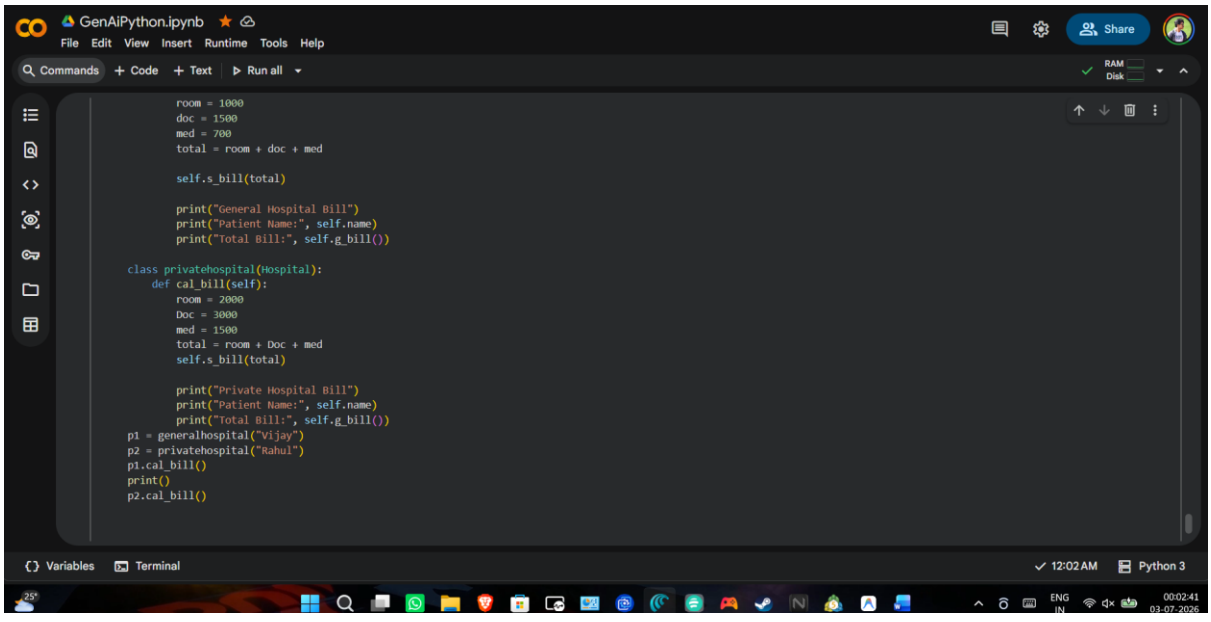


5. hospital billing with 4 pillers of oop,

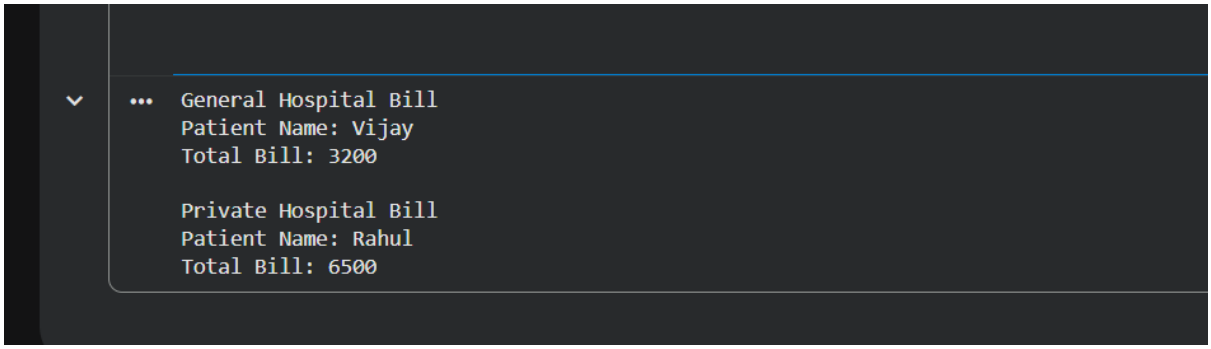
Encapsulation,Abstraction,Inheritance,Polymorphism



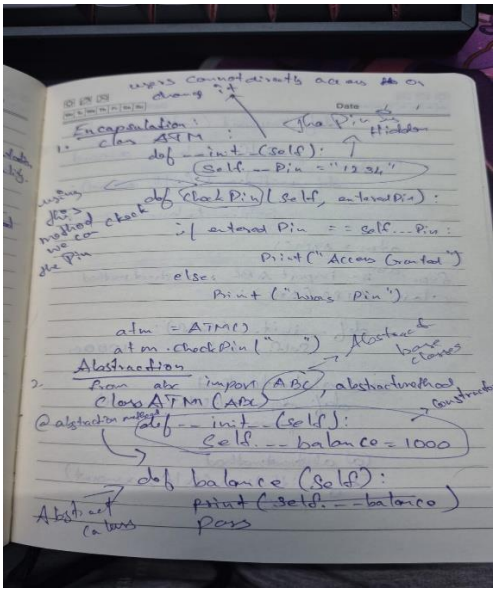
```python
from abc import ABC, abstractmethod
class Hospital(ABC):
    def __init__(self, name):
        self.name = name
        self.__bill = 0
    def s_bill(self, amount):
        self.__bill = amount
    def g_bill(self):
        return self.__bill
    @abstractmethod
    def cal_bill(self):
        pass

class generalhospital(Hospital):
    def cal_bill(self):
        room = 1000
        doc = 1500
        med = 700
        total = room + doc + med

        self.s_bill(total)

        print("General Hospital Bill")
        print("Patient Name:", self.name)
        print("Total Bill:", self.g_bill())

class privatehospital(Hospital):
    def cal_bill(self):
        room = 2000
```

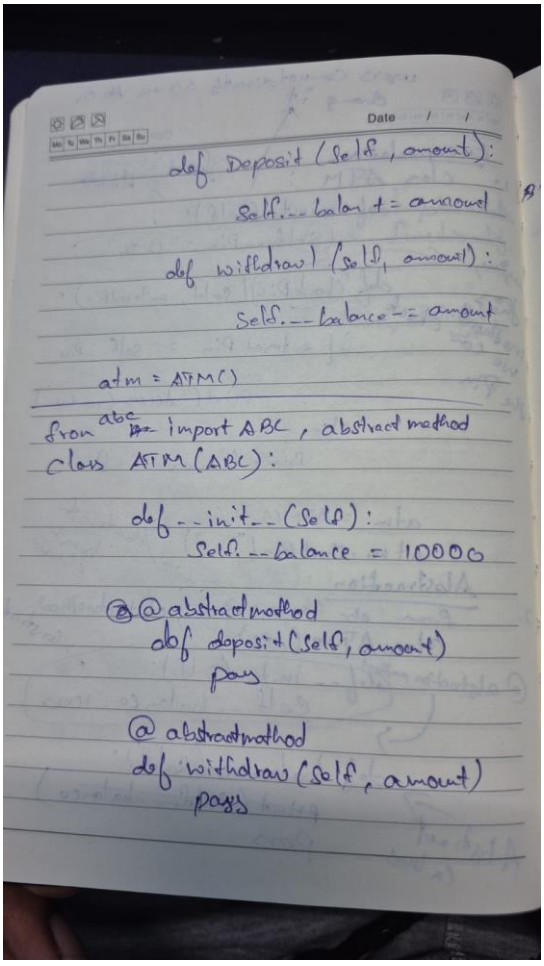


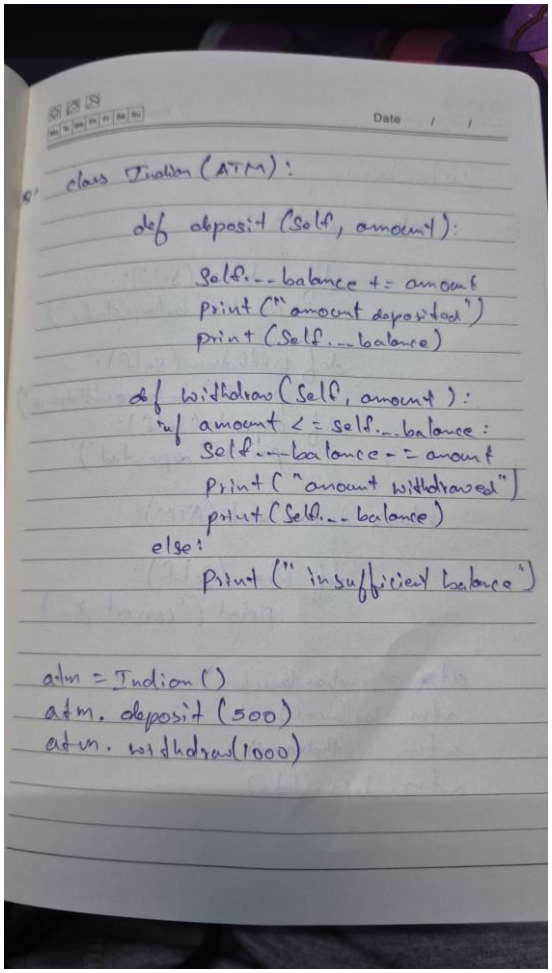
```python
        room = 1000
        doc = 1500
        med = 700
        total = room + doc + med

        self.s_bill(total)

        print("General Hospital Bill")
        print("Patient Name:", self.name)
        print("Total Bill:", self.g_bill())

class privatehospital(Hospital):
    def cal_bill(self):
        room = 2000
        Doc = 3000
        med = 1500
        total = room + Doc + med
        self.s_bill(total)

        print("Private Hospital Bill")
        print("Patient Name:", self.name)
        print("Total Bill:", self.g_bill())
p1 = generalhospital("Vijay")
p2 = privatehospital("Rahul")
p1.cal_bill()
print()
p2.cal_bill()
```



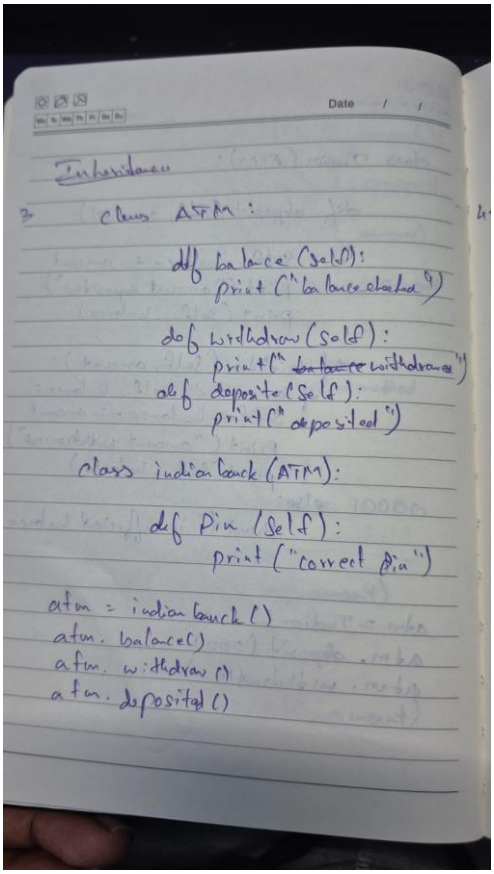
```
General Hospital Bill
Patient Name: Vijay
Total Bill: 3200

Private Hospital Bill
Patient Name: Rahul
Total Bill: 6500
```


Encapsulation:
class ATM:
def __init__(self):
self.__pin = "1234"
def checkPin(self, enteredPin):
if enteredPin == self.__pin:
Print("Access Granted")
else:
Print("Wrong Pin")
atm = ATM()
atm.checkPin("")
Abstraction
from abc import ABC, abstractmethod
class ATM(ABC):
def __init__(self):
self.__balance = 1000
def balance(self):
pass


def Deposit(self, amount):
self.__balance += amount
def withdraw(self, amount):
self.__balance -= amount
atm = ATM()
from abc import ABC, abstractmethod
class ATM(ABC):
def __init__(self):
self.__balance = 10000
@abstractmethod
def deposit(self, amount)
pass
@abstractmethod
def withdraw(self, amount)
pass

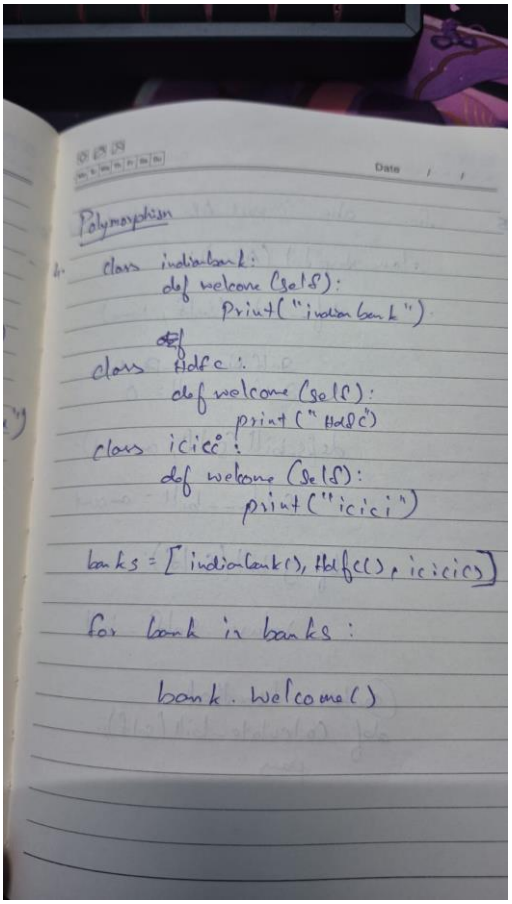


```
class Indian(ATM):
    def deposit(self, amount):
        self.__balance += amount
        print("amount deposited")
        print(self.__balance)

    def withdraw(self, amount):
        if amount <= self.__balance:
            self.__balance -= amount
            print("amount withdrawed")
            print(self.__balance)
        else:
            print("insufficient balance")

atm = Indian()
atm.deposit(500)
atm.withdraw(1000)
```



Inheritance

```
class ATM:
    def balance(self):
        print("balance checked")
    def withdraw(self):
        print("~~balance~~ withdrawed")
    def deposite(self):
        print("deposited")

class indianbank(ATM):
    def pin(self):
        print("correct pin")

atm = indianbank()
atm.balance()
atm.withdraw()
atm.deposit()
```



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Polymorphism

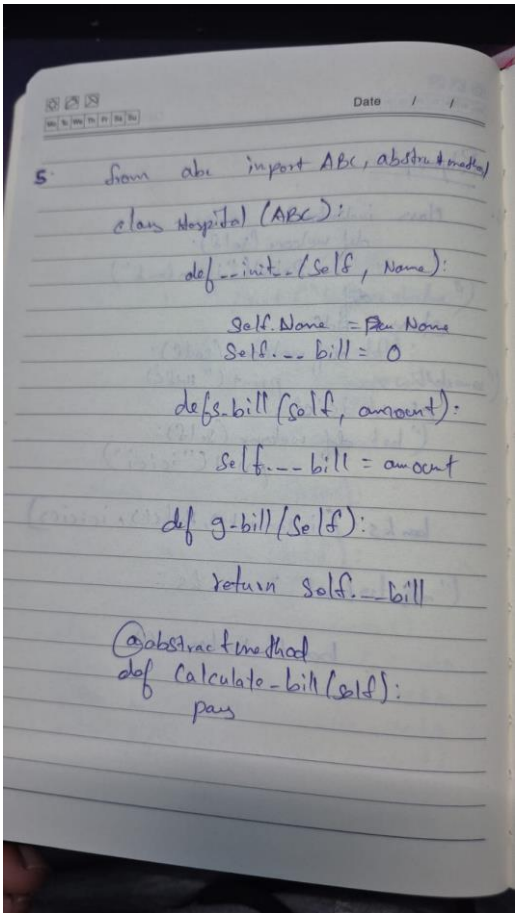
4.
```
class indianbank:
    def welcome(self):
        print("indian bank")

class hdfc:
    def welcome(self):
        print("hdfc")
class icici:
    def welcome(self):
        print("icici")

banks = [indianbank(), hdfc(), icici()]

for bank in banks:

    bank.welcome()
```



Date / /

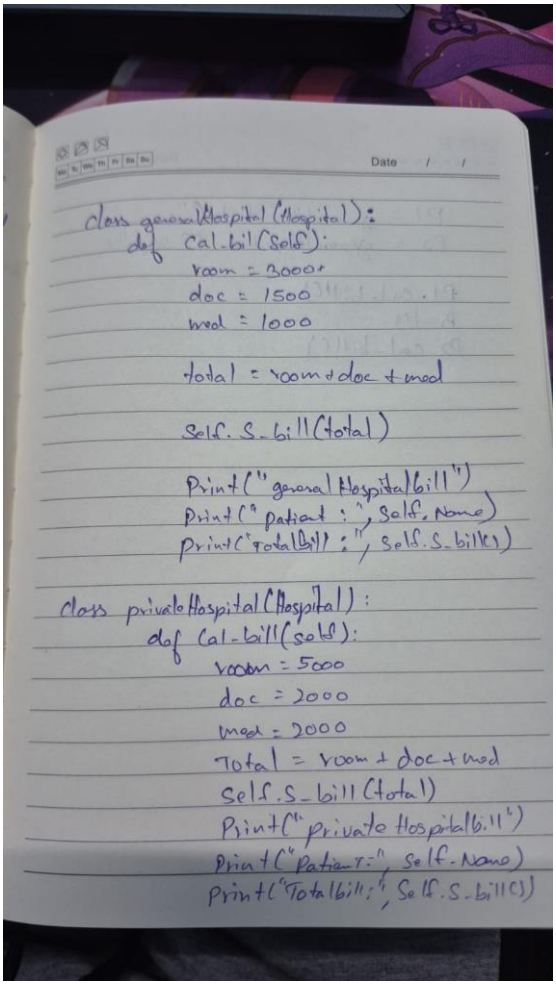
5.
```
from abc import ABC, abstractmethod

class Hospital(ABC):
    def __init__(self, Name):
        self.Name = Name
        self.__bill = 0

    def s_bill(self, amount):
        self.__bill = amount

    def g_bill(self):
        return self.__bill

    @abstractmethod
    def calculate_bill(self):
        pass
```



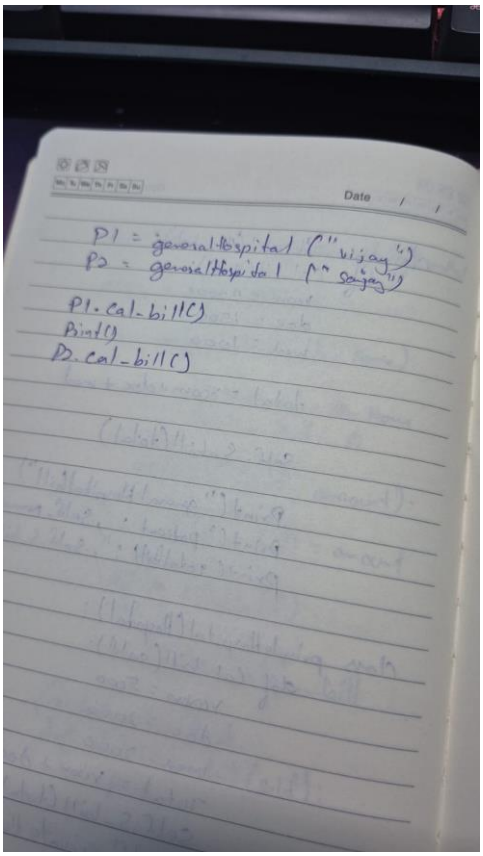
```
class generalHospital(Hospital):
    def cal_bill(self):
        room = 3000
        doc = 1500
        meal = 1000

        total = room + doc + meal

        self.s_bill(total)

        print("general Hospital bill")
        print("patient :", self.name)
        print("totalbill :", self.s_bill())

class privateHospital(Hospital):
    def cal_bill(self):
        room = 5000
        doc = 2000
        meal = 2000
        Total = room + doc + meal
        self.s_bill(total)
        print("private Hospital bill")
        print("Patient:", self.name)
        print("Total bill:", self.s_bill())
```



```
p1 = generalHospital("vijay")
p2 = generalHospital("sanjay")
p1.cal_bill()
print()
p2.cal_bill()
```